PRODUCT USER MANUAL

OMI_SI
Sea Ice Indicators
ARTIC_OMI_SI_extent
ANTARTIC_OMI_SI_Extent

Issue 5.0

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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
ANT	Antartic Ocean
ARC	Artic Ocean
SI	Sea Ice
SIE	Sea Ice Extent

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DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download. Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1. Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) ARTIC_OMI_SI_extent and ANTARTIC_OMI_SI_extent. It provides time series of Artic and Antartic Sea Ice Extent (SIE).

It contains 1D (time dimension) SIE information computed from different reanalysis data ("multi-model" approach). This product is available Northern Hemisphere and Southern Hemisphere. It is defined on a temporal axis from the 15th of January 1993. The interval between each data is a month. The ensemble mean, the spread and the single members are distributed in the NetCDF file for each time series.

ARCTIC_OMI_SI_Extent product is organised of one dataset :

- ARCTIC_OMI_SI_extent which contains monthly mean time series of Sea Ice Extent of the Northern Hemisphere.
- The product is yearly updated (i.e. temporal extension ; see product improvements pages User Notification Service | CMEMS (copernicus.eu)).

ANTARCTIC_OMI_SI_Extent product is organised of one dataset :

- ANTARCTIC_OMI_SI_extent which contains monthly mean time series of Sea Ice Extent of the Southern Hemisphere.
- The product is yearly updated (i.e. temporal extension ; see product improvements pages User Notification Service | CMEMS (copernicus.eu)).

The product is published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

Products ARCTIC_OMI_SI_extent and ANTARCTIC_OMI_SI_extent are described in the Quality Information Document (QUID) CMEMS-OMI-QUID-ARC-SIE-GREP.pdf (CMEMS-OMI-QUID-ARC-SI-extent.pdf (copernicus.eu)) and CMEMS-OMI-QUID-ANT-SIE-GREP.pdf (CMEMS-OMI-QUID-ANT-SI-extent.pdf (copernicus.eu)).

I.2. History of changes

2018/02	Publication of ARCTIC_OMI_SI_Extent and ANTARCTIC_OMI_SI_Extent product
2018/09	Update of ARCTIC_OMI_SI_Extent and ANTARCTIC_OMI_SI_Extent products (with 2017 data).
2020/04	Update of ARCTIC_OMI_SI_Extent and ANTARCTIC_OMI_SI_Extent products (with 2018 data).
2022/03	Update of ARCTIC_OMI_SI_Extent and ANTARCTIC_OMI_SI_Extent products (with 2019 data).
2024/11	Full redelievery of ARCTIC_OMI_SI_Extent and ANTARCTIC_OMI_SI_Extent products (until year 2023).

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product(s) name	ARTIC_OMI_SI ANTARTIC_OMI_SI
Geographical coverage	Arctic Ocean (Northern Hemisphere) Antarctic Ocean (Southern Hemisphere)
Variables	Sea Ice Extent (SIE)
Update frequency	Yearly
Available time series	From 1993-01-15 to regularly updated (see product improvements pages User Notification Service | CMEMS (copernicus.eu)
Delivery mechanism	MDS
Temporal/Spatial resolution	Monthly averages
Sea Ice threshold	At least 15% of sea ice in a cell
Format	NetCDF CF1.7

Detailed information on the systems and products are on CMEMS web site: CMEMS-OMI-QUID-ARC-SI-extent.pdf (copernicus.eu) and CMEMS-OMI-QUID-ANT-SI-extent.pdf (copernicus.eu).

II.2. Details of datasets

ARTIC_OMI_SI_Extent contains the 1D monthly Sea Ice Extent time series: for each time series, GLOR, CGLO, ORAS reanalysis, the ensemble mean and the corresponding standard deviation are included.
siextentn_glor [km2] Sea Ice Extent from GLORYS2V4 reanalysis sea_ice_extent **siextentn_cglo [km2]** Sea Ice Extent from C-GLORS reanalysis sea_ice_extent **siextentn_oras [km2]** Sea Ice Extent from ORAS5 reanalysis sea_ice_extent **siextentn_mean [km2]** Ensemble Mean of Sea Ice Extent from GREP reanalysis sea_ice_extent **siextentn_std [km2]** Standard Deviation of Sea Ice Extent from GREP reanalysis sea_ice_extent
ANTARTIC_OMI_SI_Extent contains the 1D monthly Sea Ice Extent time series: for each time series, GLOR, CGLO, ORAS reanalysis, the ensemble mean and the corresponding standard deviation are included.
siextentn_glor [km2] Sea Ice Extent from GLORYS2V4 reanalysis sea_ice_extent **siextentn_cglo [km2]** Sea Ice Extent from C-GLORS reanalysis sea_ice_extent **siextentn_oras [km2]** Sea Ice Extent from ORAS5 reanalysis sea_ice_extent **siextentn_mean [km2]** Ensemble Mean of Sea Ice Extent from GREP reanalysis

sea_ice_extent **siextentn_std [km2]** Standard Deviation of Sea Ice Extent from GREP reanalysis sea_ice_extent

II.3. Product System Description

As these products aim to be part of a "multi-model" approach, they provide the time series of monthly averaged SIE for each reanalysis which are used to compute the mean and the spread. The product is available for Northern hemisphere and Southern hemisphere.

The monthly mean time series start on January 1993 and is regularly updated.

III File Format

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

III FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

OMI_SI files nomenclature when downloaded through the CMEMS MDS Services is based on the following nomenclature:

<datasetname>_<timestep>_P<productiontime>.nc

Where :

- **<datasetname>** is the dataset reference **(artic_omi_sie_extent** or **antartic_omi_si_extent).**
- **<timestep>** is the first time step contained in the file.
- **<productiontime>** is the production time when the file was generated.
- **.nc** is the standard NetCDF filename extension.

Example: arctic_omi_si_extent_19930115_P20200410.nc

IV REFERENCES

See the Quality Information Document (QUID) and Product User Manual (PUM):
CMEMS-OMI-QUID-ARC-SI-extent.pdf (copernicus.eu)
CMEMS-OMI-QUID-ANT-SI-extent.pdf (copernicus.eu).